Mathematical Methods for Finance

Prof. Daniele Angelini

University of Perugia

Academic Year 2026/2027

Module 3 - Lesson 10

Random vectors and conditional methods

Joint laws, covariance, conditioning and projections

Learning outcomes for Lesson 10

By the end of the lesson students should be able to

1. read joint, marginal and conditional distributions;
2. compute covariance and correlation;
3. use Fubini-type reasoning for expectations;
4. understand conditional expectation as best information-based prediction;
5. connect conditional laws, Bayes formula, Radon–Nikodym ideas and LSMC.

How today is organised

Block 1

Joint distributions, marginals, independence, covariance and correlation.

Block 2

Conditional distributions, conditional expectation and Bayes/Radon–Nikodym intuition.

Block 3

Guided exercises and exam-style computations.

Block 1

Random vectors and joint laws

Random vectors

Definition: Random vector

A random vector is a vector of random variables:

$$X = (X_1, \ldots, X_d) : \Omega \to \mathbb{R}^d.$$

Example: Two variables

(X, Y) may describe two measurements from the same outcome. The key question is not only each distribution, but how they move together.

Joint law versus marginals

Definition: Joint law

The joint distribution of (X, Y) describes probabilities such as

$$\mathbb{P}(X \in A,\ Y \in B).$$

The marginal law of X is obtained by ignoring Y.

Counterexample: Common mistake

Knowing the two marginal distributions does not determine the joint distribution.

Discrete joint table

Example: Joint distribution

	$Y=0$	$Y=1$	$\mathbb{P}_X$
$X=0$	0.20	0.10	0.30
$X=1$	0.30	0.40	0.70
$\mathbb{P}_Y$	0.50	0.50	1

Exercise: Read the table $\star$

Compute $\mathbb{P}(X=1)$, $\mathbb{P}(Y=1)$, $\mathbb{P}(X=1, Y=1)$, $\mathbb{P}(X+Y \geq 1)$.

Solution: read the table

Worked example: Solution

$$\mathbb{P}(X = 1) = 0.70, \quad \mathbb{P}(Y = 1) = 0.50, \quad \mathbb{P}(X = 1, Y = 1) = 0.40.$$

The event $X + Y \geq 1$ excludes only $(0, 0)$, so

$$\mathbb{P}(X + Y \geq 1) = 1 - 0.20 = 0.80.$$

Marginal distributions

Definition: Marginalization

For a discrete pair,

$$\mathbb{P}(X = x) = \sum_{y} \mathbb{P}(X = x, Y = y),$$

$$\mathbb{P}(Y = y) = \sum_{x} \mathbb{P}(X = x, Y = y).$$

Intuition: Summing out

To find the law of one variable, add over all possible values of the other variable.

Independence of random variables

Definition: Independence

Random variables X and Y are independent if

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$$

for all relevant sets A, B.

Exercise: Table check $\star\star$

Using the previous table, are X and Y independent?

Worked example: Solution

Check one cell: $\mathbb{P}(X = 1, Y = 1) = 0.40$, while $\mathbb{P}(X = 1)\mathbb{P}(Y = 1) = 0.70 \cdot 0.50 = 0.35$. Not independent.

Joint densities

Definition: Joint density

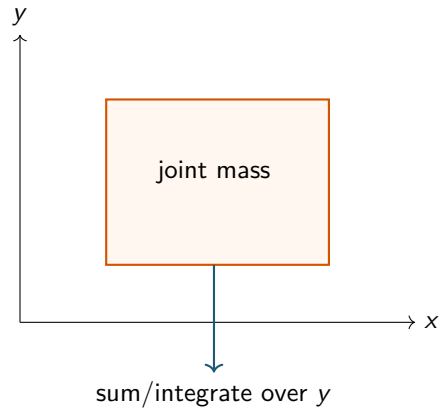
For continuous variables, a joint density $f_{X,Y}$ satisfies

$$\mathbb{P}((X, Y) \in A) = \iint_A f_{X,Y}(x, y)\, dx\, dy.$$

Marginals are obtained by integration:

$$f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x, y)\, dy.$$

Visual: marginalization as projection



Message

A marginal distribution is a projection of the joint distribution.

Fubini expectation rule

Theorem: Fubini principle

When integrability is satisfied,

$$\mathbb{E}[g(X,Y)] = \iint g(x,y) f_{X,Y}(x,y)\, dx\, dy.$$

For discrete variables,

$$\mathbb{E}[g(X,Y)] = \sum_x \sum_y g(x,y)\mathbb{P}(X=x, Y=y).$$

Exercise: Quick use $\star\star$

From the table, compute $\mathbb{E}[XY]$.

Solution: compute $\mathbb{E}[XY]$

Worked example: Discrete expectation

Only the cell $(X, Y) = (1, 1)$ contributes to XY. Therefore

$$\mathbb{E}[XY] = 1 \cdot 1 \cdot \mathbb{P}(X = 1, Y = 1) = 0.40.$$

Method

If the function $g(X, Y)$ is simple, exploit zeros before computing.

Covariance

Definition: Covariance

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Intuition: Meaning

Covariance measures signed co-movement. Positive covariance means that high values tend to appear together.

Worked covariance from a table

Exercise: Continue the previous table $\star\star\star$

Compute $\mathrm{Cov}(X, Y)$.

Worked example: Solution

$\mathbb{E}[X] = \mathbb{P}(X = 1) = 0.70$, $\mathbb{E}[Y] = \mathbb{P}(Y = 1) = 0.50$, and $\mathbb{E}[XY] = 0.40$. Hence

$$\mathrm{Cov}(X, Y) = 0.40 - 0.70 \cdot 0.50 = 0.05.$$

Correlation

Definition: Correlation

If $\mathrm{Var}(X) > 0$ and $\mathrm{Var}(Y) > 0$,

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}}.$$

Properties

$$-1 \leq \mathrm{Corr}(X, Y) \leq 1.$$

Correlation is scale-free: it has no physical unit.

Uncorrelated does not mean independent

Counterexample: Classical example

Let $U \sim \text{Unif}(-1, 1)$, $X = U$, $Y = U^2$. Then

$$\text{Cov}(X, Y) = \mathbb{E}[U^3] - \mathbb{E}[U]\mathbb{E}[U^2] = 0.$$

But X and Y are not independent because Y is determined by X.

Intuition: Message

Covariance detects linear dependence, not every form of dependence.

Variance of linear combinations

Theorem: Two variables

$$\operatorname{Var}(aX + bY) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) + 2ab \operatorname{Cov}(X, Y).$$

If X, Y are independent, then $\operatorname{Cov}(X, Y) = 0$, provided second moments exist.

Exercise: Quick use $\star\star$

If $\operatorname{Var}(X) = 4$, $\operatorname{Var}(Y) = 9$, $\operatorname{Cov}(X, Y) = 1$, compute $\operatorname{Var}(X + 2Y)$.

Solution: linear combination

Worked example: Calculation

Using $a = 1$, $b = 2$:

$$\mathrm{Var}(X + 2Y) = 1^2 \cdot 4 + 2^2 \cdot 9 + 2 \cdot 1 \cdot 2 \cdot 1 = 4 + 36 + 4 = 44.$$

Counterexample: Common mistake

Do not forget the covariance term unless independence or zero covariance is known.

Cholesky decomposition

Motivation

A covariance matrix Σ encodes variances and covariances. To simulate a vector with this covariance, one often writes

$$\Sigma = LL^{\top}.$$

Then $X = LZ$, with Z standard independent components, has covariance Σ.

Example: Two-dimensional case

For

$$\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix},$$

one factor is

$$L = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{pmatrix}.$$

Block 1 mini-workshop

Exercise: Three tasks

1. From a 2×2 joint table, compute both marginals.
2. Check independence by comparing one cell with the product of marginals.
3. Compute covariance using $\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.

Target skill

Read a joint distribution table without losing track of the objects.

Block 2

Conditional distributions and conditional expectation

Conditional distribution in a table

Definition: Discrete conditional law

If $\mathbb{P}(Y = y) > 0$,

$$\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}.$$

Exercise: Use the previous table $\star\star$

Compute the conditional law of X given $Y = 1$.

Worked example: Solution

$\mathbb{P}(Y = 1) = 0.50$. Thus $\mathbb{P}(X = 0 \mid Y = 1) = 0.10/0.50 = 0.20$, $\mathbb{P}(X = 1 \mid Y = 1) = 0.40/0.50 = 0.80$.

Conditional expectation from a conditional law

Definition: Discrete conditional expectation

$$\mathbb{E}[X \mid Y = y] = \sum_x x \mathbb{P}(X = x \mid Y = y).$$

Exercise: Continue $\star\star$

Using the conditional law of $X \mid Y = 1$, compute $\mathbb{E}[X \mid Y = 1]$.

Worked example: Solution

$$\mathbb{E}[X \mid Y = 1] = 0 \cdot 0.20 + 1 \cdot 0.80 = 0.80.$$

Conditional expectation as a random variable

Definition: Conditional expectation

$\mathbb{E}[X \mid Y]$ is a random variable depending on the information contained in Y.

Intuition: Prediction

It is the best prediction of X using only the information in Y, in mean-square sense.

Counterexample: Common mistake

$\mathbb{E}[X \mid Y]$ is not usually a number. It becomes a number only after fixing a value of Y.

Tower property

Theorem: Tower property

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X].$$

More generally, conditioning then averaging recovers the unconditional expectation.

Intuition: Meaning

Predictions made with partial information are still unbiased on average.

Conditional expectation rules

Rules to remember

- If X is determined by Y, then $\mathbb{E}[X \mid Y] = X$.
- If X is independent of Y, then $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$.
- Linearity holds: $\mathbb{E}[aX + bZ \mid Y] = a\mathbb{E}[X \mid Y] + b\mathbb{E}[Z \mid Y]$.

Exercise: Quick classification $\star$

If X and Y are independent and $\mathbb{E}[X] = 2$, what is $\mathbb{E}[X \mid Y]$?

Solution: independence and conditioning

Worked example: Answer

If X is independent of Y, knowing Y gives no information about X. Therefore

$$\mathbb{E}[X \mid Y] = \mathbb{E}[X] = 2.$$

Object check

The expression $\mathbb{E}[X \mid Y]$ is still a random variable, but here it is the constant random variable equal to 2.

Law of total expectation

Theorem: Partition version

If $A_1, \dots, A_n$ form a partition, then

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X \mid A_i]\mathbb{P}(A_i).$$

Exercise: Two groups $\star\star$

A variable has conditional means 10 in group A and 4 in group A^c. If $\mathbb{P}(A) = 0.3$, compute $\mathbb{E}[X]$.

Worked example: Solution

$\mathbb{E}[X] = 10(0.3) + 4(0.7) = 5.8$.

Bayes as conditional reweighting

Intuition: Reweighting

Bayes formula changes weights after observing information. The prior weight $\mathbb{P}(A)$ becomes the posterior weight $\mathbb{P}(A \mid B)$.

Formula

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Exercise: Interpret $\star$

Which factor is the likelihood: $\mathbb{P}(A)$, $\mathbb{P}(B \mid A)$, or $\mathbb{P}(B)$?

Radon–Nikodym intuition

Idea

When probabilities are described by densities, conditional probabilities and expectations can be understood as normalized weights.

Definition: Density viewpoint

If a probability measure has density f, then probabilities are computed by

$$\mathbb{P}(A) = \int_A f(x)\,dx.$$

Intuition: Connection

Radon–Nikodym derivatives generalize the idea of density to abstract spaces.

Least Square Monte Carlo: idea only

Projection viewpoint

Conditional expectation can be approximated by projecting onto a finite-dimensional space of functions.

Example: Regression approximation

Approximate

$$\mathbb{E}[Y \mid X] \approx a_0 + a_1 X + a_2 X^2.$$

The coefficients are selected by least squares.

Intuition: Connection to Module 1

This is a projection idea in an L^2-space.

Block 2 mini-workshop

Exercise: Three conditional tasks

1. Compute $\mathbb{P}(X = x \mid Y = y)$ from a table.
2. Compute $\mathbb{E}[X \mid Y = y]$ from the conditional law.
3. Decide when $\mathbb{E}[X \mid Y]$ is constant.

Target skill

Conditioning means changing the information and renormalizing probabilities.

Block 3

Exam-style exercises

Guided exercise 1: joint table

Exercise: Problem $\star\star$

	$Y=-1$	$Y=1$
$X=0$	0.10	0.20
$X=2$	0.30	0.40

Compute the marginal distributions of X and Y, and check independence.

Solution: guided exercise 1

Worked example: Detailed solution

$$\mathbb{P}(X = 0) = 0.30, \quad \mathbb{P}(X = 2) = 0.70.$$

$$\mathbb{P}(Y = -1) = 0.40, \quad \mathbb{P}(Y = 1) = 0.60.$$

Check $(X = 0, Y = -1)$: $0.10 \neq 0.30 \cdot 0.40 = 0.12$. Hence not independent.

Guided exercise 2: covariance

Exercise: Problem $\star\star\star$

Using the same table, compute $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathbb{E}[XY]$ and $\mathrm{Cov}(X, Y)$.

Solution: guided exercise 2

Worked example: Detailed solution

$$\mathbb{E}[X] = 0(0.30) + 2(0.70) = 1.4.$$

$$\mathbb{E}[Y] = -1(0.40) + 1(0.60) = 0.2.$$

$$\mathbb{E}[XY] = 0 + 2(-1)(0.30) + 2(1)(0.40) = -0.6 + 0.8 = 0.2.$$

$$\mathrm{Cov}(X, Y) = 0.2 - 1.4 \cdot 0.2 = -0.08.$$

Guided exercise 3: conditional expectation

Exercise: Problem $\star\star\star$

Using the same table, compute $\mathbb{E}[X \mid Y = -1]$, $\mathbb{E}[X \mid Y = 1]$, and verify the tower property.

Solution: guided exercise 3

Worked example: Conditional means

Given $Y = -1$, total probability is 0.40:

$$\mathbb{P}(X = 2 \mid Y = -1) = 0.30/0.40 = 0.75,$$

so $\mathbb{E}[X \mid Y = -1] = 2(0.75) = 1.5$. Given $Y = 1$, $\mathbb{P}(X = 2 \mid Y = 1) = 0.40/0.60 = 2/3$, so $\mathbb{E}[X \mid Y = 1] = 4/3$. Then

$$1.5(0.40) + (4/3)(0.60) = 0.6 + 0.8 = 1.4 = \mathbb{E}[X].$$

Independent exercise

Exercise: Try alone $\star\star\star$

Let X, Y have joint probabilities

	$Y=0$	$Y=2$
$X=-1$	0.25	0.15
$X=1$	0.25	0.35

Compute $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathbb{E}[XY]$, $\mathrm{Cov}(X,Y)$, and $\mathbb{E}[Y \mid X=1]$.

Solution: independent exercise

Worked example: Solution

$\mathbb{P}(X = -1) = 0.40$, $\mathbb{P}(X = 1) = 0.60$, so $\mathbb{E}[X] = 0.20$. Also $\mathbb{P}(Y = 0) = 0.50$, $\mathbb{P}(Y = 2) = 0.50$, so $\mathbb{E}[Y] = 1$.

$$\mathbb{E}[XY] = (-1)2(0.15) + 1 \cdot 2(0.35) = -0.30 + 0.70 = 0.40.$$

$$\mathrm{Cov}(X, Y) = 0.40 - 0.20 \cdot 1 = 0.20.$$

Given $X = 1$, $\mathbb{P}(Y = 2 \mid X = 1) = 0.35/0.60 = 7/12$. Thus $\mathbb{E}[Y \mid X = 1] = 2(7/12) = 7/6$.

Exit ticket

Exercise: Three questions

1. What is the difference between a marginal and a conditional distribution?
2. Does zero covariance imply independence?
3. Why is $\mathbb{E}[X \mid Y]$ a random variable?

End-of-lesson checklist

You should now be able to

- read a joint distribution table;
- compute marginals and conditionals;
- check independence from a joint law;
- compute covariance and correlation;
- use the tower property;
- interpret conditional expectation as information-based prediction.

Suggested practice after Lesson 10

Core practice

From the Module 3 exercise pack: random vectors, joint distributions, marginal distributions, conditional probability, covariance and conditional expectation.

Exercise: Minimum training set

Choose at least:

- two joint-table exercises;
- two covariance/correlation exercises;
- two conditional expectation exercises;
- one Bayes exercise in a table format.